

Data Modeling (Part 2)

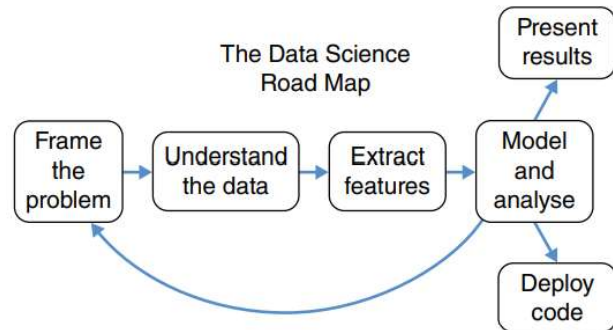
Le Ngoc Thanh
lnthanh@fit.hcmus.edu.vn
Department of Computer Science

Ho Chi Minh City

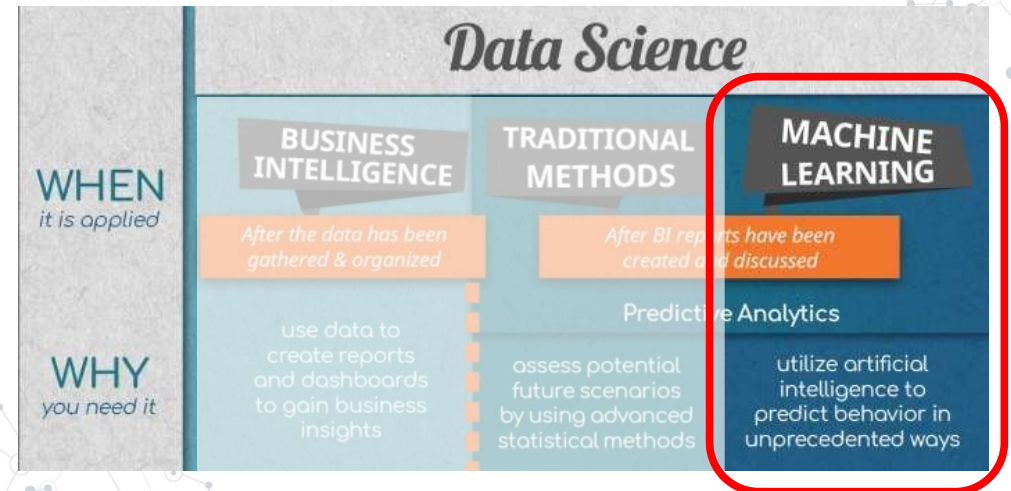
Contents

- Data science and machine learning review
- Classification model
- Clustering model

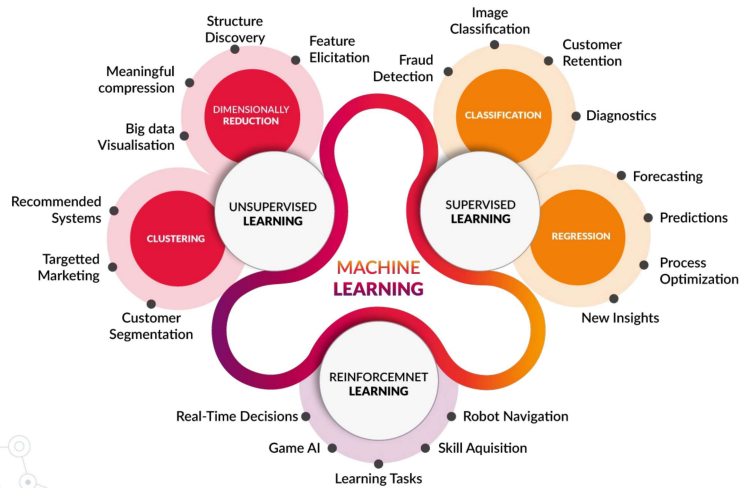
Process



Data Science's tasks



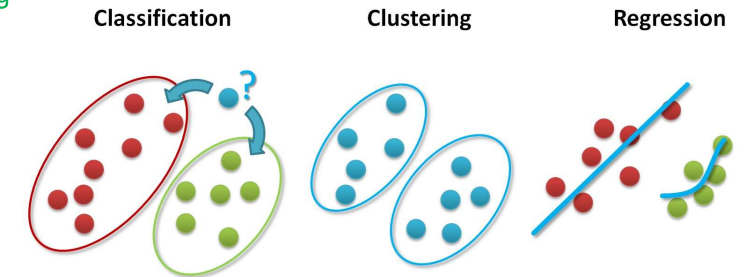
ML Tasks



The course's focus

◎ In this course, we focus on **three main groups** of ML:

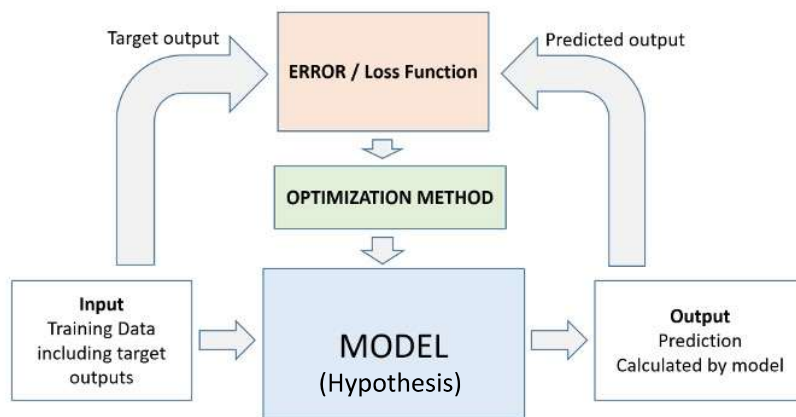
- Regression
- Classification
- Clustering



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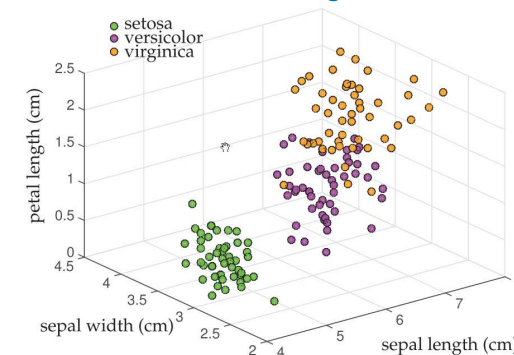
General model learning architecture



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Classification

◎ **Classification** is the problem of identifying which of a set of categories an observation belongs to.



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Classification

- ◎ The inputs and outputs for the learning binary classification task can be stated as follows:

- Input

data $\{\mathbf{x}_j \in \mathbb{R}^n, j \in Z := \{1, 2, \dots, m\}\}$

labels $\{\mathbf{y}_j \in \{\pm 1\}, j \in Z' \subset Z\}$

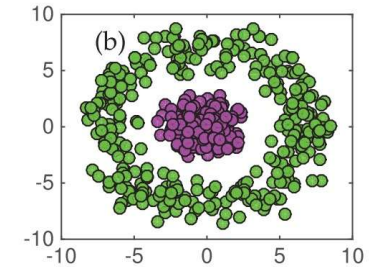
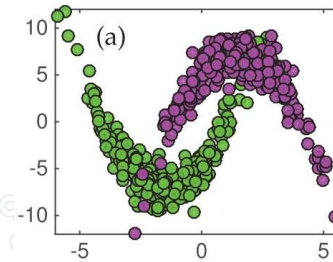
- Output

labels $\{\mathbf{y}_j \in \{\pm 1\}, j \in Z\}$

Challenges

- ◎ Some challenges in classification task:

- The **boundary** between the data forms a **nonlinear manifold** that is difficult to characterize.
- If the **sampling data only captures a portion** of the manifold, then it will almost surely fail in characterizing population data.
- Data can be in **higher dimensional space** and **visualization** is essentially **impossible**.



Well-known classification algorithms

- ◎ Some well-known classifier:

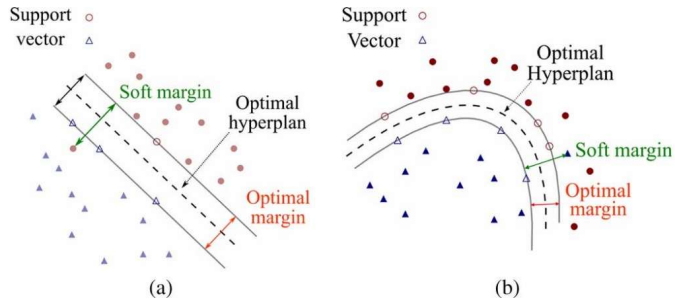
- Support Vector Machine (SVM)
- Classification and Regression Tree (CART)
- k-nearest Neighbors (kNN)
- Naïve Bayes
- Ensemble Learning and Boosting (AdaBoost)
- Ensemble Learning of Decision Tree (C4.5)
- Deep neural nets

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 - **Neural network**
- ◎ **Clustering model**

Support Vector Machines

- ◎ The original SVM algorithm by Vapnik and Chervonenkis evolved out of the statistical learning literature in 1963, where **hyperplanes are optimized to split the data** into distinct classes.



SVM classifier scheme. (a) Linear and (b) nonlinear cases.

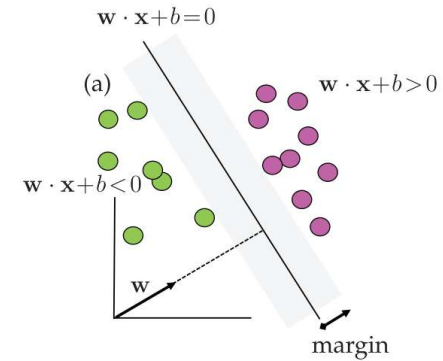
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Linear SVM

- ◎ The key idea of the linear SVM method is to construct a hyperplane:

$$w \cdot x + b = 0$$

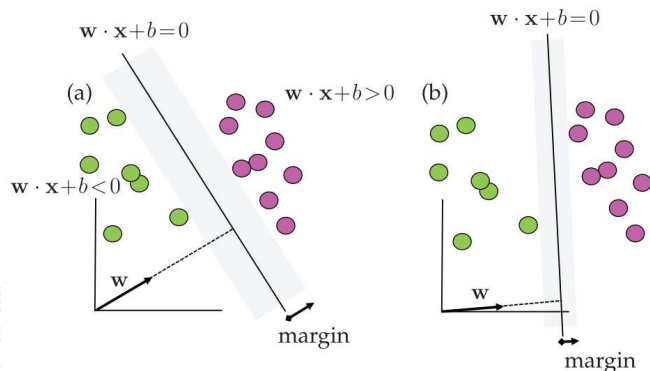
where the vector w and constant b parametrize the hyperplane.



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Optimization problem in SVM

- ◎ The optimization problem in SVM includes:
 - Optimize a decision line which makes **the fewest labeling errors**.
 - Optimizes **the largest margin** between the data.



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Objective function

- ◎ The loss function is defined as follows:

$$L(y_j, \hat{y}_j) = L(y_j, \text{sign}(w \cdot x_j + b))$$

$$= \begin{cases} 0 & \text{if data is correctly labeled} \\ +1 & \text{if data is incorrectly labeled} \end{cases}$$

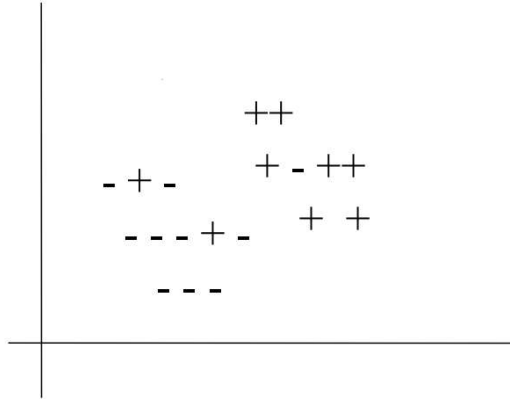
- ◎ The goal is also to make the margin as large as possible:

$$\text{argmin}_{w,b} \sum_{j=1}^m L(y_j, \hat{y}_j) + \frac{1}{2} \|w\|^2 \text{ subject to } \min_j |w \cdot x_j + b| = 1$$

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Noisy/Nonlinear Classification

- How to separate the following data by SVM?



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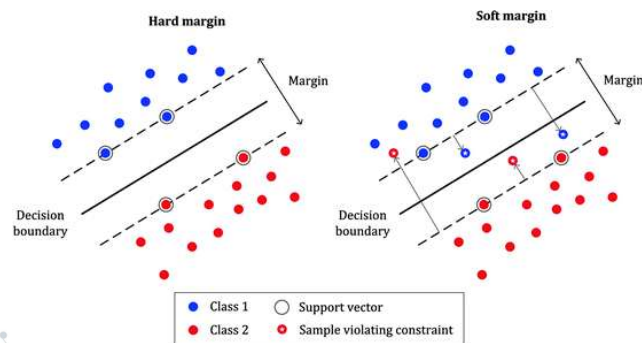
Noisy/Nonlinear Classification

- Two basic approaches:
 - Use a linear classifier, but allow some (penalized) errors
 - Soft margin, slack variables
 - Project data into higher dimensional space
 - Do linear classification there
 - Kernel functions

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Soft margin

- Margin violation means choosing an hyperplane, which can allow some data points to stay in either in between the margin area or in the incorrect side of hyperplane.



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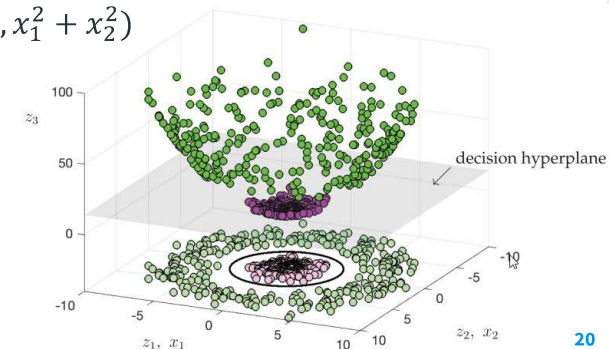
Map to higher-dimensional space

- Embedding the data in a higher dimensional space:

$$\mathbf{x} \rightarrow \Phi(\mathbf{x})$$

- For example:

$$(x_1, x_2) \rightarrow (z_1, z_2, z_3) := (x_1, x_2, x_1^2 + x_2^2)$$



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Kernel trick

- ◎ **Kernel function** (similarity function) takes input vectors in the original space and return dot product of these vectors in the feature space (a real number).

$$K(\mathbf{x}, \mathbf{z}) = \Phi(\mathbf{x}) \cdot \Phi(\mathbf{z})$$

Kernel trick

- ◎ The **objective function** only includes the **dot product of the transformed feature vectors**.

$$\mathbf{w} = \sum_{j=1}^m \alpha_j \Phi(\mathbf{x}_j)$$

$$f(\mathbf{x}) = \mathbf{w} \cdot \Phi(\mathbf{x}) + b = \sum_{j=1}^m \alpha_j \Phi(\mathbf{x}_j) \cdot \Phi(\mathbf{x}) = \sum_{j=1}^m \alpha_j K(\mathbf{x}_j, \mathbf{x})$$

- ◎ Therefore, just substitute these dot product terms with the kernel function, and don't even use $\Phi(\mathbf{x})$.

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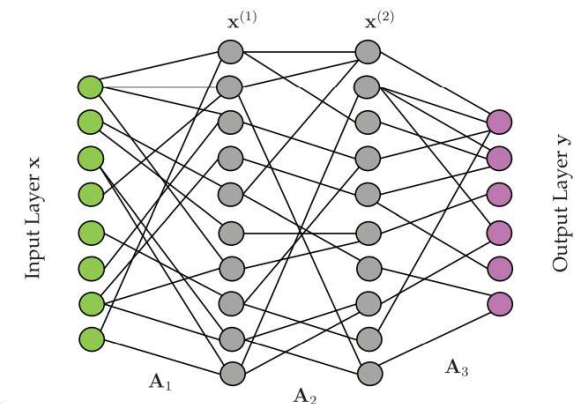
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 - **Neural network**
- ◎ Clustering model

Generic architecture of a multi-layer NN

- ◎ For classification tasks, the goal of the NN is to map a set of input data to a classification.

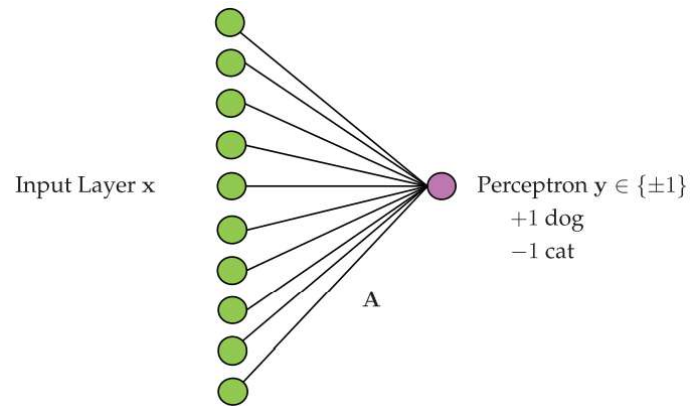


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One-layer network

- First, consider a single layer network for binary classification:



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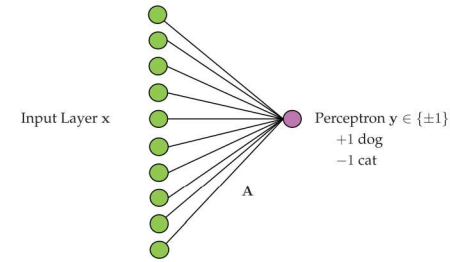
One-layer network

- Hypothesis with linear mapping:

$$AX = Y$$

$$\rightarrow [a_1 \ a_2 \ \dots \ a_n] \begin{bmatrix} | & | & | & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_p \\ | & | & | & | \end{bmatrix} = [+1 \ +1 \ \dots \ -1 \ -1]$$

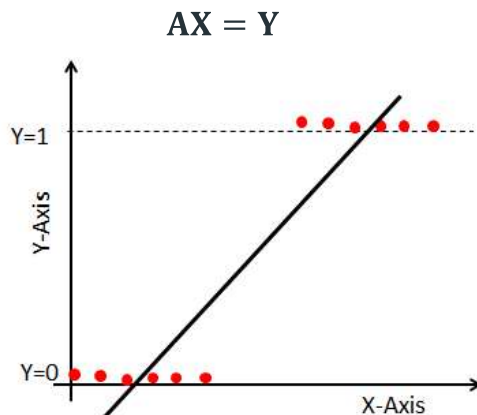
where each column of the matrix X is a dog or cat image ($\mathbf{x}_j \in \mathbb{R}^n$) and the columns of Y are its corresponding labels.



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One-layer network

- The linear mapping:



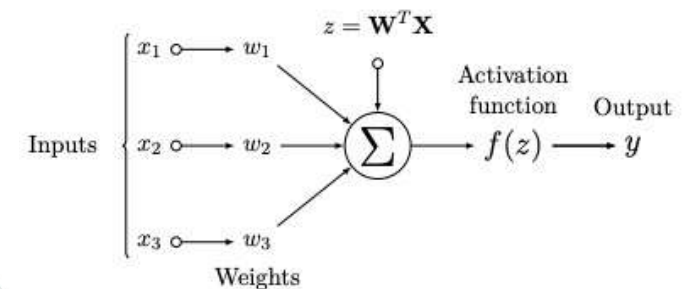
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Nonlinear transformations

- Hypothesis with nonlinear mapping:

$$y = f(A, x)$$

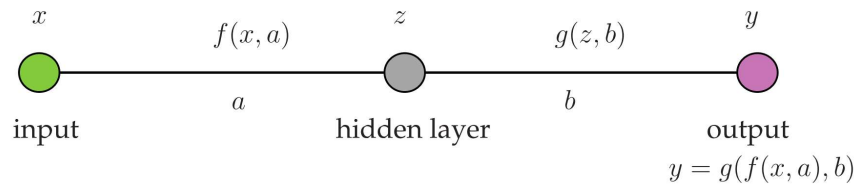
where $f(\cdot)$ called an **activation function** (transfer function) in neural networks.



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Neural network optimization

- ◎ The **optimization of NN** (determine the weights of the network) is done through the **backpropagation process**.
 - The process forces the optimization to **backpropagate error** through the network relies on the **chain rule of differentiation**.
- ◎ For example, with simple neural network:



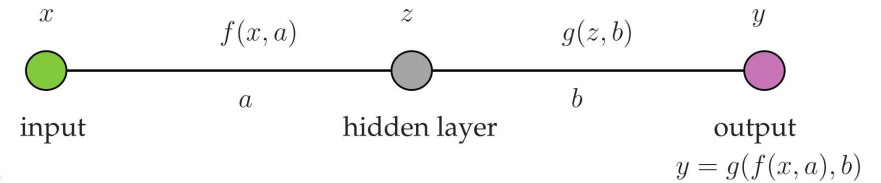
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Neural network optimization

- ◎ The compositional structure is:

$$y = g(z, b) = g(f(x, a), b)$$
- ◎ The error function is:

$$E = \frac{1}{2}(y_0 - y)^2$$



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Neural network optimization

- ◎ Partial derivative:

$$\frac{\partial E}{\partial a} = -(y_0 - y) \frac{\partial y}{\partial b} = -(y_0 - y) \frac{\partial y}{\partial z} \frac{\partial z}{\partial a} \quad (\text{chain rule})$$

$$\frac{\partial E}{\partial b} = -(y_0 - y) \frac{\partial y}{\partial b}$$

- ◎ Gradient descent:

$$a_{k+1} = a_k + \eta \frac{\partial E}{\partial a_k}$$

$$b_{k+1} = b_k + \eta \frac{\partial E}{\partial b_k}$$

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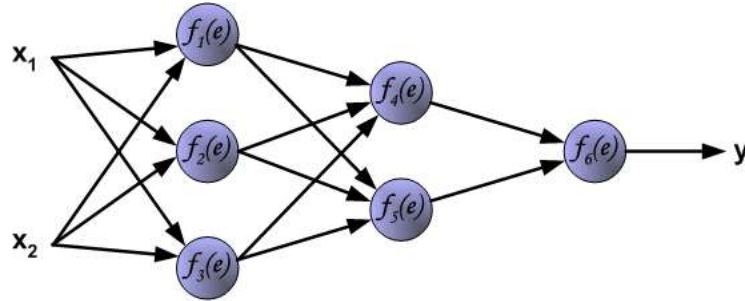
Overall progress of neural network

1. A NN is specified along with a labeled training set.
2. The initial weights of the network are set to random values.
3. The training data is run through the network to produce an output y , whose ideal ground-truth output is y_0 .
4. The derivatives with respect to each network weight is then computed using backprop formulas.
5. For a given learning rate η , the network weights are updated as in gradient descent equation.
6. Return to step (3) and continue iterating until a maximum number of iterations is reached or convergence is achieved.

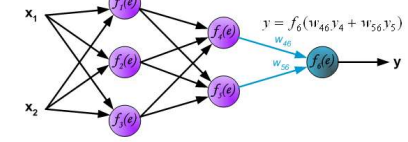
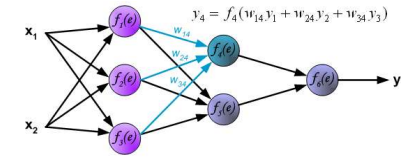
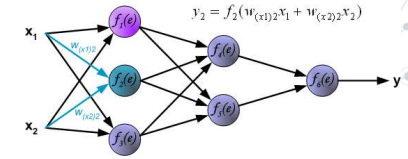
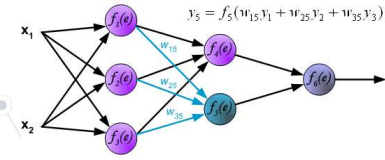
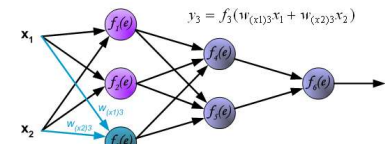
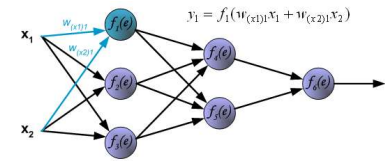
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Overall progress of neural network

1. A NN is specified along with a labeled training set.
2. The initial weights of the network are set to random values.



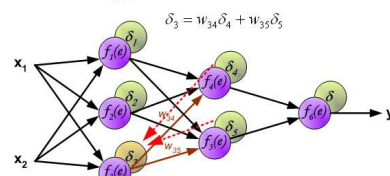
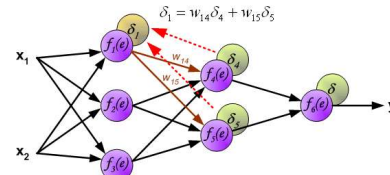
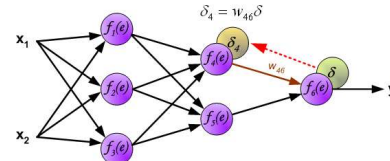
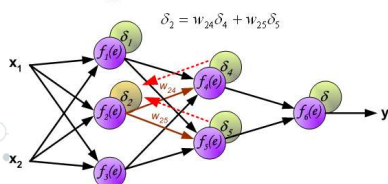
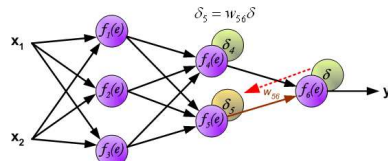
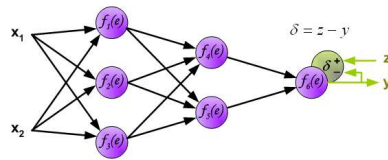
3. The training data is run through the network



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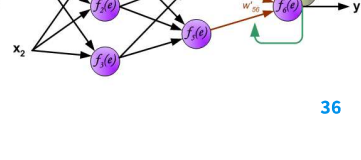
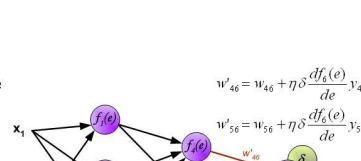
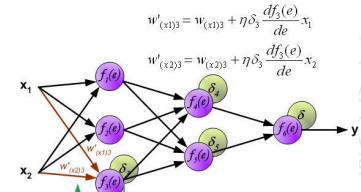
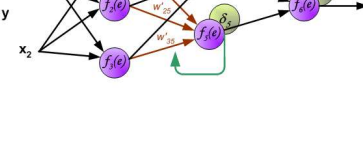
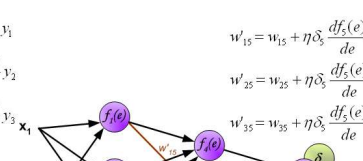
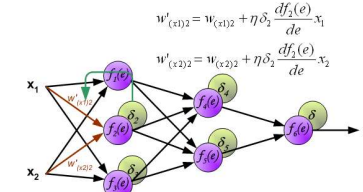
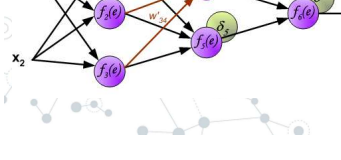
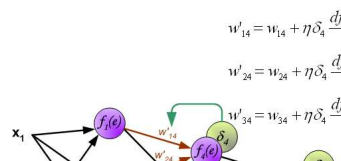
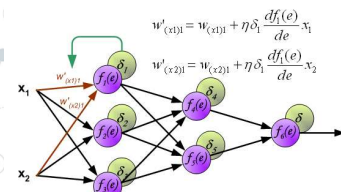
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Backpropagate error



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4. The derivatives with respect to each network weight is computed.
5. The network weights are updated as in gradient descent equation.

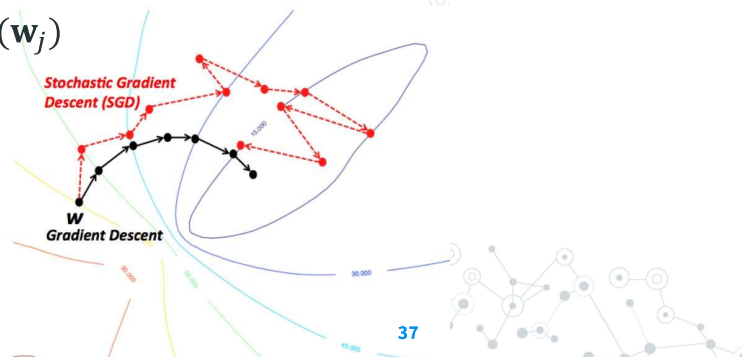


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Stochastic Gradient Descent

- Stochastic Gradient Descent (SGD): **a single, randomly data point (k)** is chosen to approximate the gradient at each step of the iteration.

$$\mathbf{w}_{j+1} = \mathbf{w}_j - \eta \nabla E_k(\mathbf{w}_j)$$

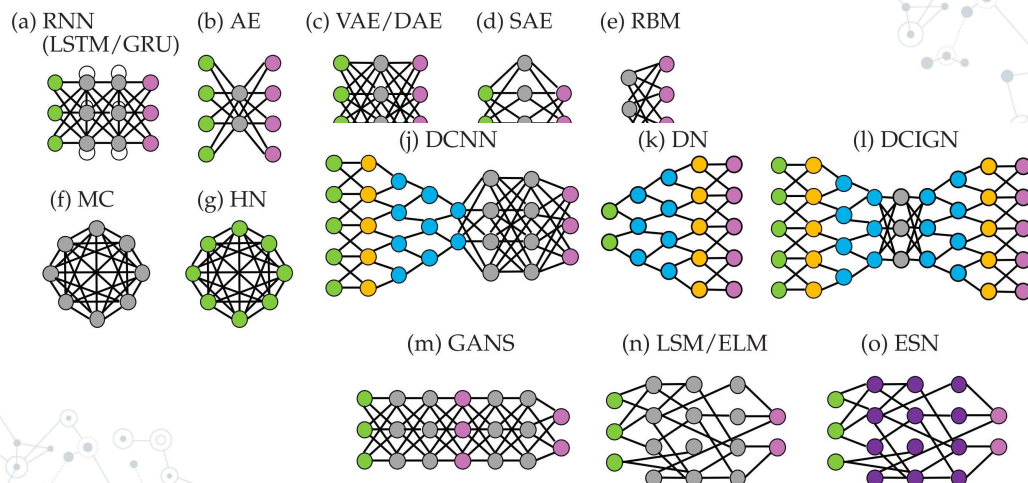


Batch gradient descent

- If instead of a single point, a subset of points (K) is used, then we have the following **batch gradient descent** algorithm.

$$\mathbf{w}_{j+1} = \mathbf{w}_j - \eta \nabla E_K(\mathbf{w}_j)$$

Deep Learning



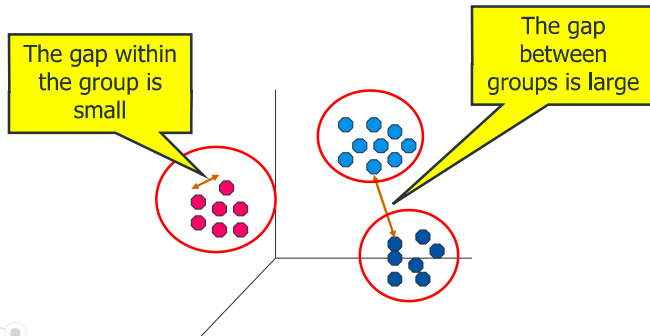
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Clustering

Clustering is the process of grouping objects into clusters:

- The same group objects are highly similar.
- And very different from the subject in the rest of the groups.

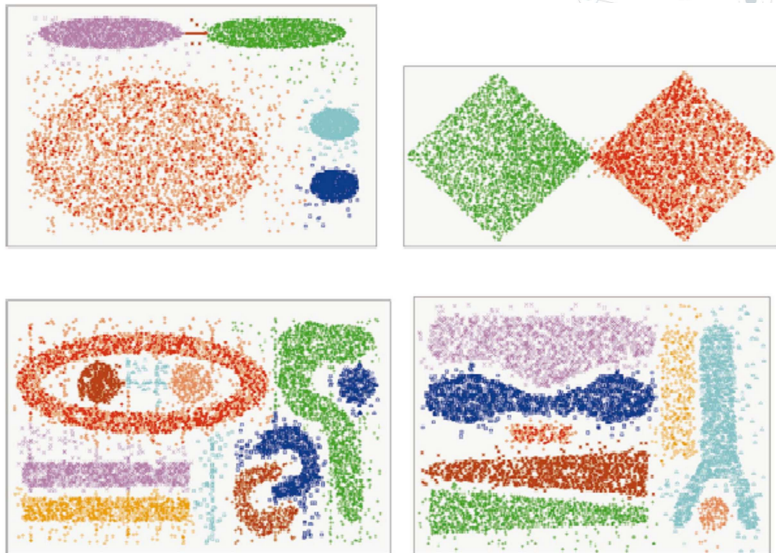


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Clustering

- Clustering is an unsupervised learning because the label/class is not pre-defined
- Therefore, clustering is a type of learning by visual rather than learning by examples

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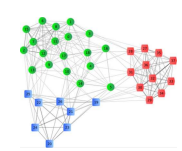


CHAMELEON

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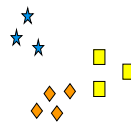
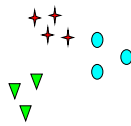
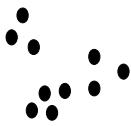
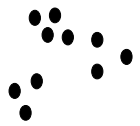
Some applications of clustering

- Grouping of related documents for web browsing
- Groups of genes and proteins have the same function
- Group of stocks with the same volatility
- Groups of areas of the same land type in geography
- Identify homegroups by Home type, value, and geographic location
- Defining a Gaming object group



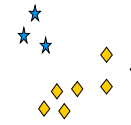
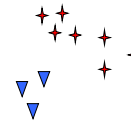
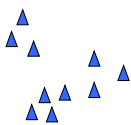
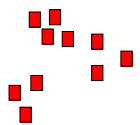
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What is a group?



How many groups are there?

6 groups



2 groups

4 groups

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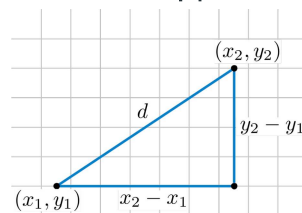
A good Clustering?

- ⦿ A good clustering method will have to create groups of high quality:
 - The same level in the High group.
 - Similar levels to other low-level groups.
- ⦿ The quality of the clustering depends on:
 - Analog measurement
 - Its enforcement
 - Ability to discover some or all of the underlying patterns

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Measure Similarity

- ⦿ The distance used is mainly to measure the same or not similar level between the two objects.
 - Examples: Euclidean gap, Cosin Gap, Minkowski, Mahattan...
- ⦿ Distance functions differ in value ranges, types, ranks, and variable elements.
- ⦿ The weights of the dependent variables on the application and the data implications.



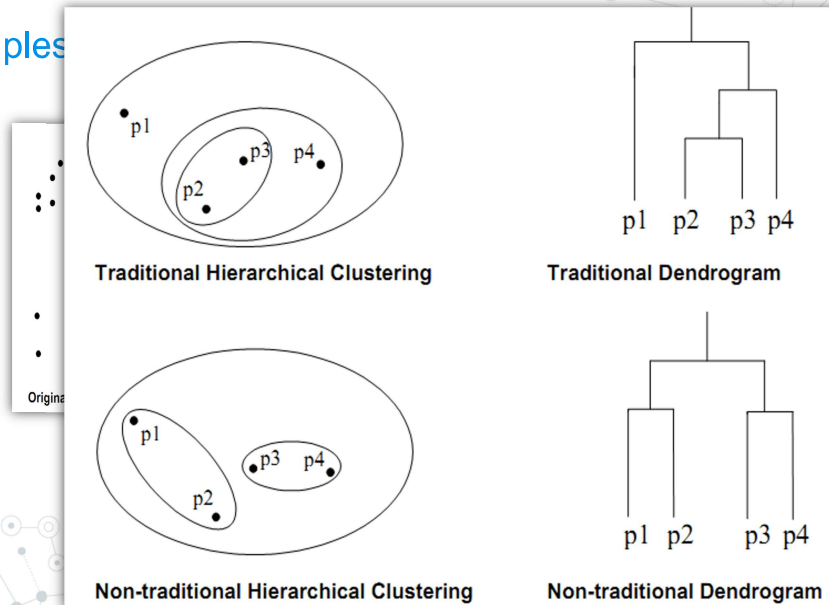
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Some clustering methods

- ⦿ Partitional clustering
 - Formation of partitions and evaluate them based on some criteria
 - Algorithms: K-Means, K-Medoids, CLARANS
- ⦿ Hierarchical clustering
 - Create the division of Layers
 - Algorithms: Diana, Agnes, BIRCH, CAMELEON
- ⦿ Density-based clustering
 - Based on the jaw and density function
 - Algorithms: DBSACN, OPTICS, DenClue

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Examples



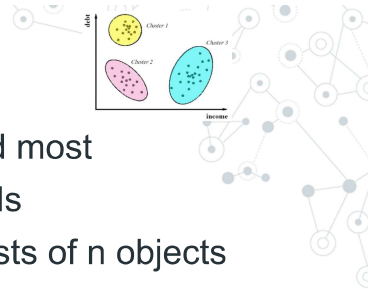
Some clustering methods (2/2)

- ◎ Grid-based methods
 - Based on multi-level particle structure
 - Algorithms: STING, WaveCluster, CLIQUE
- ◎ Model-based methodology
- ◎ Frequent pattern-based methods
- ◎ Methods based on binding or user guidance
- ◎ Link-based method

...

Partitioning Clustering

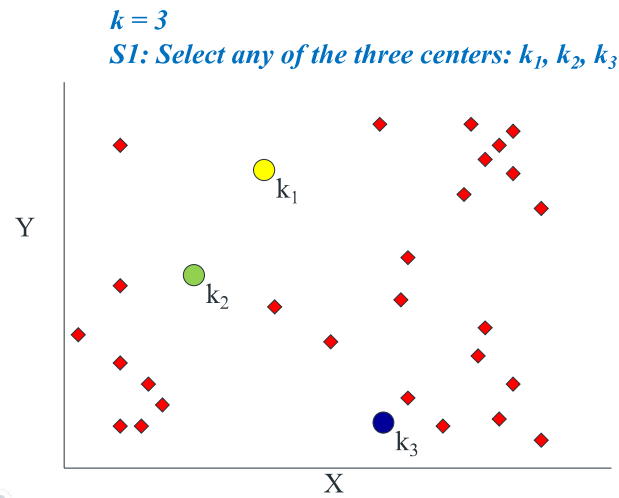
- ◎ Partitioning Clustering Is the simplest and most foundational method of clustering methods
- ◎ Idea: The partition of a D database consists of n objects into k Groups so that it optimizes the criteria of the partition.
- ◎ Global optimization: Full expression of all groups
- ◎ Heuristic algorithm:
 - K-means: each group is performed by the center value of the group
 - K-Medoids, PAM: each group is performed by one of the groups ' objects



K-Means algorithm (1/2)

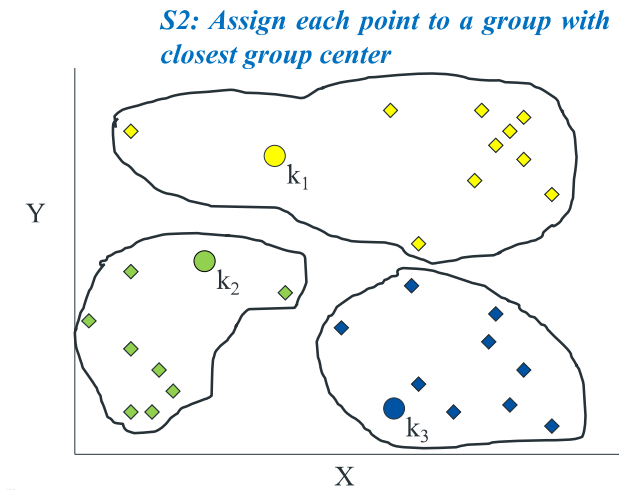
- ◎ Before number k, each group is performed by the group's centroid value.
 - S1: Select Random K objects as the center of the groups
 - S2: Assigns each remaining object to the closest group based on distance measurement such as Euclidean, Cosine analogue, correlation,...
 - S3: Calculating the center value of each group based on newly joined objects.
 - S4: If the group Center has nothing to change or there are only a few points that change the group, stop, back to S2.

K-means example (1/4)



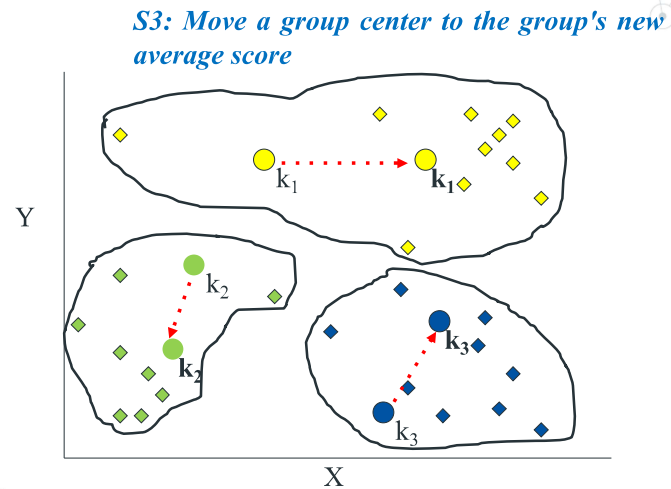
53

K-means example (2/4)



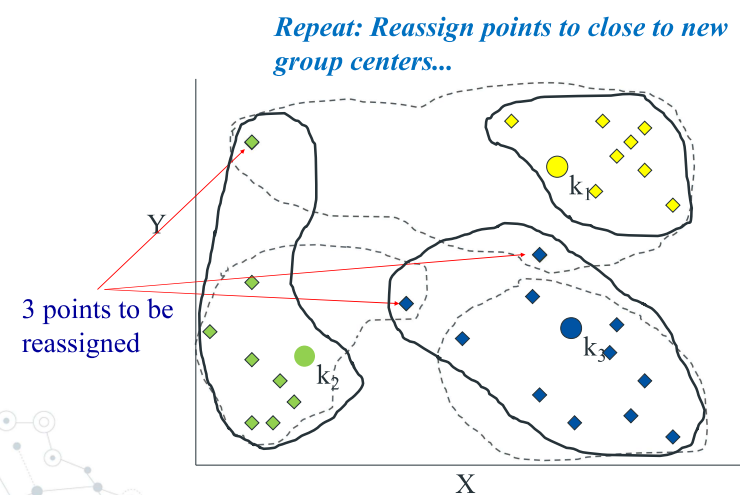
54

K-means example (3/4)



55

K-means example (4/4)



56

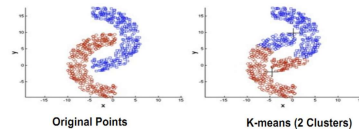
k-means

◎ Pros:

- Simple, effective. Complexity $O(TKN)$; $T, K \ll N$.
Achieve local optimization.

◎ Cons:

- Only available to objects in a continuous n-dimensional space
Need to determine the number K group before
Sensitivity to noise and personal data
Not suitable for exploring groups with circles/bridges



Exercise 1

- Use the K-means algorithm and Euclidean spacing to Clustering 8 samples into 3 groups:

$A1=(2,10)$, $A2=(2,5)$, $A3=(8,4)$, $A4=(5,8)$, $A5=(7,5)$, $A6=(6,4)$, $A7=(1,2)$, $A8=(4,9)$.

Distance matrix based on Euclidean is given in the following slide.

Assuming the initializing seed (center) is $k1 = A1$, $k2 = A4$ and $k3 = A7$. Run K-Means 1 time. Identify groups to be formed. Where is the new center of each group

- Draw in Space 10 x 10 The samples were bundled with the group over 1 run (draw frames) and the center of each group. How many K-means loops will converge (stop)? Illustrate the results at each loop.

The End